THURSDAY
October 29, 2026

DELHI
CITY EDITION
22 Pages ₹ 12.00

THE HINDU

www.thehindu.com

INDIA'S NATIONAL NEWSPAPER SINCE 1878

Vol. 16 • No. 256
Regd. DL(ND)-11/6110/2006-07-08
RNI No. UPENG/2012/49940

To subscribe,
give a missed call at
9828331787
or scan QR code

Printed at » Chennai » Coimbatore » Bengaluru » Hyderabad » Madurai » Noida » Visakhapatnam » Thiruvananthapuram » Kochi » Vijayawada » Mangaluru » Tiruchirapalli » Kolkata » Hubballi » Mohali » Malappuram » Mumbai » Lucknow » Cuttack » Patna

SC asks if poll panel has powers to punish voters

ECI in 'error' to force pre-existing voters whose names were deleted from rolls to apply as new electors, says SC; it asks how electors can be denied vote because they were lax or inert during SIR

Krishnadas Rajagopal
NEW DELHI



"Of the 1,45,10,299 electors in Delhi at the start of the SIR, 47,56,722 were marked as having not returned their enumeration forms"

PRASHANT BHUSHAN
Counsel for petitioners



"Let us say a voter was lax or did not respond; will that be a ground for you [ECI]... to forfeit his or her right to vote for ever?"

JUSTICE JOYMALYA BAGCHI
Supreme Court judge

The Supreme Court on Wednesday questioned the authority of Election Commission of India (ECI) to deprive citizens forever of their right to vote simply because they were lax in returning enumeration forms during the special intensive revision (SIR) of electoral rolls, which has disenfranchised nearly 13 crore voters across the country.

"Let us say a voter was lax, inert, or did not respond, will that be a ground for you, the Election Commission of India, to forfeit his or her right to vote for ever? We will answer that," Justice Joymalya Bagchi, part of the Bench headed by Chief Justice of India Surya Kant, addressed the poll body. The court said the ECI was in "error" to force pre-existing voters, whose names were deleted from the electoral roll, to apply as new voters under Form 6 of the Registration of Electors Rules, 1960.

Senior advocate Dama Seshadri Naidu, for the ECI, said the court's Bihar SIR judgment on May 27 had mentioned that the poll body had "residuary powers".

Justice Bagchi clarified that the Bihar judgment was not a "binding precedent" and asked why the ECI had not then used these residuary powers to reach out to voters who had not responded to enumeration forms.

CONTINUED ON
» PAGE 12
SIR: SERIOUS QUESTIONS
» PAGE 8

64.5 lakh electors 'missing' in Karnataka's rolls

Around 64.5 lakh electors are potentially missing from the final electoral rolls of Karnataka released after the special intensive revision on Tuesday, as per calculations based on Elector-to-Population ratio of 66.11, released by the office of the Chief Electoral Officer. » PAGE 9

Hunker down

Proactive governance kept the casualties low after Cyclone Arnab

Cyclone Arnab, the first named storm of the North Indian Ocean cyclone season this year, intensified rapidly before making landfall as a 'very severe cyclonic storm' on October 28. This particular rapidity is a challenge for forecasters and some models did fail to anticipate how strong Arnab would become. It also moved faster than expected towards the coast, before stalling for a time and beginning the landfall process several hours before the India Meteorological Department (IMD) had predicted it would, in the middle of the night, leaving many district authorities with a potentially narrower window to evacuate high-risk settlements. With core wind speeds of 125-135 kilometres per hour, it delivered more than 100 mm of rain over parts of Anakapalli, Visakhapatnam, Kalingapatnam, Kakinada and Machilipatnam, in Andhra Pradesh. But even though it has completed landfall, southern Odisha likely faces a greater threat only hence. Early reports claimed no loss of life in Andhra Pradesh or Odisha, which has been attributed to the disaster management operations in both States; the Andhra Pradesh Chief Minister, N. Chandrababu Naidu, said there were five deaths in Andhra Pradesh. Regrettably, the suspension of transport services, including road connectivity and flights, closure of schools and offices, and power cuts to minimise risks posed by loose wires remains an essential part of saving lives, entailing other social and economic disruptions that the States must account for, and address, in their post-storm assessments.

Odisha may have to brace for rainfall and waters due to moisture in the storm's outer circulation, which will arrive over land that has recovered in patches from the fourth wettest monsoon in the State's history. Extreme rainfall after repeated flooding may face soil that has not drained fully, raising the risk of more flooding following even lower levels of rainfall, and slope failures in hilly terrain. Farmers in Ganjam and Puri have already voiced worries about the risk of prolonged inundation destroying paddy crops, compounding the already considerable financial consequences of the southwest monsoon; Ganjam alone had lost 33% of crops by October. In Sundargarh, however, farmers hope that Arnab will relieve moisture stress in standing crops after a dry month. The IMD has already issued alerts in the Ganjam, Gajapati, Rayagada, Koraput, Malkangiri, Nabarangpur, and Kalahandi districts, before the rainbelt extends further north. The relatively low figure of deaths will be a testament to proactive governance. However, it is past time that States adopted a zero-disruption approach, especially since property and infrastructure were damaged, and the risk of people being pushed into poverty over lost possessions and livelihoods persists.

The SIR verdict, serious constitutional questions



Bishwajit Bhattacharyya

Senior Advocate, Supreme Court of India, and a former Additional Solicitor General of India

The top court must ensure fairness in electoral roll revisions

Did the Supreme Court of India err in its verdict dated May 27, 2026, on the Special Intensive Revision (SIR) of electoral rolls by empowering the Election Commission of India (ECI) to undertake a "limited" inquiry into citizenship for the inclusion of names in, or their exclusion from, the electoral rolls?

Second, can crystallised electoral rolls be unsettled in subsequent elections by deleting names on suspicion?

These questions arise in the context of the electoral rolls in Bihar, published on January 6, 2025, following the ECI's Special Summary Revision (SSR).

On June 24, 2025, the ECI decided to undertake a fresh SIR, beginning with Bihar. Thus, every electoral roll in Bihar between January 1, 2003 (the date of the last SIR in Bihar), and January 6, 2025 (the date of the last SSR, in Bihar), was, in effect, called into question by the ECI. This was unnecessary, as the ECI, in any case, periodically updates the electoral rolls efficiently.

This unusual exercise effectively called into question the electoral rolls used in every election held in Bihar between 2003 and 2025.

Consequently, in Bihar, the SSR dated January 6, 2025, was superseded by the SIR ordered on June 24, 2025.

The ECI's notification dated June 24, 2025, was challenged before the Court. However, the challenge was unsuccessful, with the Court endorsing the ECI's stand in its verdict dated May 27, 2026.

Even before the Court delivered its verdict on May 27, 2026, Assembly elections were held in Bihar in November 2025 and West Bengal in April 2026, following the ECI's SIR exercise, which entailed mass deletions and surprising inclusions of voters. Widespread dissatisfaction prevailed.

The SIR is now being implemented in other States amid protests.

The ECI's powers

The Court's verdict clarified that the ECI has not been empowered to determine citizenship in the "strict sense". The Court held that the power of the ECI affects an "individual's entitlement to be included in the electoral roll". It also held: "It does not, however, operate to divest the individual of claims of citizenship, nor does it foreclose a determination of that question by the competent authority under the Citizenship Act."

With respect, the words "limited" and "strict sense", in the context of an inquiry into citizenship, carry no legal significance, even as *obiter dicta* of the Court.

And the statement that the "power of ECI" "affects" an "individual's entitlement" to be "included in the electoral roll" is not capable of any interpretation other than that the ECI has the power to disenfranchise an existing voter.

This seems to give *carte blanche* to the ECI. Little wonder, then, that the ECI has been proceeding to alter the existing electoral rolls nonchalantly.

Enquiry means enquiry, no less; citizenship means citizenship, no less. The ECI's constitutional authority to inquire into citizenship remains contentious. Even a limited inquiry risks exceeding the ECI's electoral mandate. The Court seems to have erred.

While preparing the electoral roll, the ECI's function is limited only to ensuring that every person included is an adult Indian citizen.

The minimum age (18 years) and Indian citizenship are required to be ensured through documents. The ECI's role is strictly ministerial; it has no authority to conduct any inquiry – limited or otherwise – into citizenship. The problem, however, arises because no single document conclusively proves citizenship.

In other words, the Court vested the ECI with unguided discretion to remove a name from an existing electoral roll on the grounds of suspicious Indian citizenship. This amounts to a citizen, whose name was already on the electoral roll as of January 6, 2025, and who was presumed to be an Indian citizen, being disenfranchised by the ECI through the SIR process with effect from July 1, 2025.

Documents, citizenship, deletion

The Court's verdict did not even specify any document that could serve as proof of Indian citizenship.

Significantly, however, paragraph 158 of the Court's verdict reads, *inter alia*: "... it may not be out of place to note that a Ration Card, unlike a Passport or Birth Certificate, is certainly not a conclusive proof of citizenship..." Thus, the Court has ruled that a passport or birth certificate is conclusive proof of citizenship.

Is this why the Ministry of External Affairs, during a press briefing marking Passport Seva Divas on June 24, 2026, made the controversial statement that a passport was merely a travel document and not proof of Indian citizenship?

Could the Court not have, in the exercise of its jurisdiction, specified the documents that prove Indian citizenship? Or, could the Court not have issued a writ of mandamus directing the government to specify the documents that prove Indian citizenship?

The Court did neither, belying the hopes of every Indian citizen whose citizenship is questioned by the ECI.

How does one prove one's Indian citizenship? This question still cannot be answered clearly, yet the ECI can disenfranchise a person who has already voted in the last election merely on the grounds of doubts about their Indian citizenship.

On what basis have electoral rolls in India been prepared since January 26, 1950? How can an electoral roll used in a concluded election be altered on the grounds of suspicion? How can a person be disenfranchised after having voted in the previous election? What is the constitutional benchmark for disenfranchising an existing voter? These are questions that the Court could have considered.

The Supreme Court's powers are extensive under Article 32 (the right to constitutional remedies and the Court's power to issue writs for the enforcement of fundamental rights), Article 141 (the binding nature of the law declared by the Court on all courts in India), Article 142 (the power to pass orders necessary to do complete justice in any cause or matter pending before it), and Article 144 (the obligation of all civil and judicial authorities in India to act in aid of the Court). The Court can surely still set right this legal vacuum by clearly laying down a yardstick for establishing Indian citizenship.

Adulthood and Indian citizenship are the only two – repeat, only two – qualifications required for inclusion in the electoral roll. The Constitution does not prescribe any third mandatory qualification.

The question of disqualification arises only after identifying qualified voters. Only a person who is otherwise qualified can be disqualified.

Article 326 specifies the grounds for disqualification: "non-residence", "unsoundness of mind", "crime", or "corrupt or illegal practice".

The Representation of the People Act, 1950, and the Registration of Electors Rules, 1960, must comply with Article 326 of the Constitution. The overriding principles of natural justice must also be followed when disqualifying a qualified voter.

The name of a qualified voter simply cannot be removed from the electoral roll without a pre-decisional hearing.

Protecting franchise

Free and fair elections are part of the basic structure of India's Constitution. India's potential to emerge as a world power cannot be realised if this foundation and structure is weakened. If an Indian citizen remains on tenterhooks about their right to vote, how can we expect the best from such a citizen?

Although the right to vote has generally been characterised as a statutory right, its exercise is central to representative democracy and the constitutional commitment to free and fair elections. This was metamorphosed from the Court's judgments in *I.C. Golaknath vs State of Punjab* (1967), *Kesavananda Bharati vs State of Kerala* (1973) and *Indira Nehru Gandhi vs Raj Narain* (1975), and continues to hold the field. In the interpretation of this writer, the three judgments, read together, surely elevate the voting right to a basic feature of India's Constitution.

The *Kesavananda Bharati* judgment established that even Parliament's power to amend the Constitution cannot be exercised to destroy or damage its basic structure. The principle continues to safeguard essential constitutional features, including democracy.

My Lords, uphold the right to vote freely, fairly and on time.

India-New Zealand trade ties set to scale new heights



Anant Goenka

President, The Federation of Indian Chambers of Commerce and Industry (FICCI)

The India-New Zealand FTA will strengthen trade, investment and professional mobility

In 1953, Sir Edmund Hillary of New Zealand and Tenzing Norgay, the Nepalese-Indian Sherpa mountaineer made history by becoming the first ever to climb Mount Everest. This extraordinary partnership showed what co-operation, trust and determination could achieve, and we are once again set to see the India-New Zealand partnership scale to new heights with the entry into force of the India-New Zealand Free Trade Agreement (FTA).

The India-New Zealand relationship has always been cordial and extends beyond diplomatic ties. Their people share the same passion for cricket, place a high value on domestic traditions and share strong people-to-people ties. In fact, New Zealand is also home to a large Indian diaspora, strengthening relations between the two countries, be it in business, education or cultural exchange. The implementation of the FTA provides another opportunity to leverage these ties for a stronger economic partnership.

Moreover, this FTA is among the recently signed modern FTAs of India which will strengthen its foothold and economic engagement in the Indo-Pacific. The complementary nature of the two economies will help in creating greater opportunities for both countries, not just in trade but also in investments and other areas of economic co-operation.

High potential for trade expansion

India-New Zealand bilateral merchandise trade, which is at $1.2 billion, is way below potential. India caters to just 1.9% of New Zealand's merchandise imports worth $47.5 billion, while China accounts for an over 22% share and Australia, the United States, South Korea and Japan together account for another 33% share.

Most of these major suppliers (excluding the U.S.) already enjoy preferential access to the New Zealand market through respective trade agreements, thus placing Indian exporters at a competitive disadvantage.

With the India-New Zealand FTA now in force, Indian exporters will also get a level playing field in this market across all tariff lines that have been now offered duty free access.

The immediate gains for India could be significant in the labour-intensive sectors. New Zealand imported nearly $1.2 billion worth of apparel in 2025, but India's share was only around 4%. The elimination of average tariffs of nearly 10% on apparel offers Indian apparel manufacturers and exporters an opportunity to expand their presence in New Zealand. Similar potential exists in sectors such as leather, footwear, home textiles and carpets.

The opportunity is equally important in higher-value manufacturing. New Zealand imports nearly $7 billion worth of machinery and $5.2 billion worth of vehicles and automotive components annually. Yet, India's market share in these categories remains around 1% or lower. The FTA provides an opportunity for Indian engineering and automotive manufacturers to deepen their presence in New Zealand for these products.

Greater market access

The agreement also addresses regulatory barriers, which will create an opportunity for sectors such as pharmaceuticals. India already enjoyed zero-duty access for pharmaceuticals in New Zealand and has held a share of around 5.5% in New Zealand's imports. The FTA includes provisions that facilitate the use of regulatory evaluation and inspection reports from recognised regulators. This can reduce duplicative compliance requirements and further improve market access for Indian pharmaceutical companies.

New Zealand's services imports stood at $24.2 billion in 2024, and India's share was less than 3%. The FTA has provided commitments across 118 services sectors, including IT, professional services, healthcare, education and financial services.

The agreement also creates dedicated pathways for skilled professionals, providing a quota of 5,000 visas at any given time and stay for up to three years.

Additionally, the agreement ensures enhanced student mobility and post-study work opportunities. These will strengthen educational and professional linkages that exist between the two countries.

As part of the agreement, New Zealand will facilitate $20 billion worth of foreign direct investment into India over a 15-year period. This should not be difficult, given India's attractiveness to global investors and New Zealand's capital and technological expertise. India offers significant investment opportunities across sectors, be it high-value manufacturing, renewable energy, agritech, food processing, digital technologies or infrastructure.

Effective utilisation is the key

The outcomes from the FTA will flow only if they are effectively utilised. Indian businesses need targeted market intelligence, stronger distribution networks, regulatory preparedness and commercial partnerships. What is equally important is sustained engagement with New Zealand's investors to develop a pipeline of commercially viable projects.

Through the combined efforts of all stakeholders – the government, the associations as well as the industry on both sides – it should not be difficult to double India-New Zealand bilateral trade by 2030. The agreement has created an enabling framework and success will depend on how these opportunities are converted into exports, investments, jobs and meaningful partnerships.

Disinvestment as key economic strategy



Chandrajit Banerjee

Director General, Confederation of Indian Industry

Seven months into 2026-27, the fiscal year has been shaped by global uncertainty, high energy prices and worries over a deficient monsoon. Against this backdrop, disinvestment and asset monetisation have helped in navigating a challenging fiscal year. While receipts were already on course to meet the budget target, the government's approval of the sale of its share in the IDBI Bank on October 14 makes it likely that the target will be exceeded. For the industry and investors, the receipts are a signal of the government's continued commitment to reforms and a larger role for the private sector. This year so far, the disinvestment trajectory has progressed well, even before the IDBI stake sale. According to DIPAM data, the Centre mobilised ₹62,124 crore in April-September 2026 through disinvestment and asset monetisation, of which ₹55,757 crore came from disinvestment. These collections had already reached 78% of the full-year budget target of ₹80,000 crore. With the IDBI proceeds, the collections are to head past the target.

This comes at a time when the government looks to diversify its revenue sources amid rising expenditure commitments. Net tax revenue rose by a modest 3.4% in April-August 2026 and non-tax revenue by 3.2%, while capital expenditure was up by over 18%. Overall, the fiscal deficit has already reached 42% of the budgeted levels which implies a tight balance between revenue shortfalls and spending risks.

However, disinvestment should not be seen merely as a tool for meeting fiscal targets. It is a policy choice aimed at unlocking greater value from public assets, widening public ownership, and improving efficiency and governance through greater market discipline. The 2022 LIC IPO which drew in many first-time retail investors and policyholders remains a prime example.

This was precisely the rationale behind the government's 2021 New Public Sector Enterprise (PSE) Policy which envisages a "bare minimum presence" of PSEs in strategic sectors. Likewise, the National Monetisation Pipeline (NMP) re-engines capital via mature public assets. A target of ₹16.72 lakh crore has been set under NMP 2.0 for FY26-30 after having achieved 90% of its Phase 1 goal.

Demand-based approach

This year, however, until this month, receipts had come mainly through Offer for Sale (OFS) measures which help to broaden ownership but keep the status quo on an enterprise's operations. The strategic sale of the IDBI Bank since Air India in 2022 is an encouraging sign. From an industry perspective, the IDBI sale is more than a one-off sale. It is one of the largest Foreign Direct Investment (FDI) deals in India's financial sector and constitutes as the first step towards a 'Privatisation 2.0' framework. More importantly, this framework is premised on being investor-led, professionally managed and time-bound. Two changes would help to turn into a template.

First, the selection criteria for disinvestments of PSEs should move to a demand-based approach instead of being led by the government. It would work better to first gauge interest across a wider pool and prioritise PSEs that draw strong bids at expected valuations, with structured investor feedback flagging procedural and regulatory hurdles early. Second, a rolling three-year privatisation stream, based on investor interest and the government's strategic priorities, would give investors time for due diligence, which means deeper participation and better price discovery.

Given the fact that privatisation of several PSEs is a long journey, calibrated disinvestment could serve as an interim measure

Text&Context

THE HINDU

Is GRAP enough to curb pollution?

What is GRAP, and why is it imposed in Delhi-NCR? Which factors worsen air pollution during winter? What restrictions are imposed under Stage I of GRAP? What additional curbs come into effect under Stages III and IV? Has GRAP consistently improved Delhi's winter air quality?

EXPLAINER

Nikhil M. Babu

The story so far:

As air quality in Delhi worsened to the 'poor' category in the national capital, the Commission for Air Quality Management in NCR and adjoining areas (CAQM) on last Wednesday announced Stage 1 of the Graded Response Action Plan (GRAP) in Delhi and the National Capital Region (NCR) to bring down the pollution.

Every winter, air pollution spikes in Delhi-NCR and many parts of the larger Indo-Gangetic Plains, mainly due to meteorological factors such as lower wind speeds and a drop in temperature. Pollution from stubble burning in October and November and firecrackers during the festive season adds to emissions from vehicles and dust, further worsening air quality.

What is GRAP?

The Graded Response Action Plan (GRAP) is a set of short-term emergency measures to reduce air pollution and is divided into four stages. The CAQM announces the different stages, depending on the level of pollution recorded or forecast for the near future.

Whenever the CAQM announces GRAP, it directs State governments to implement the restrictions with "immediate effect" in entire Delhi and also in the NCR districts of Haryana, Uttar Pradesh, and Rajasthan. The restrictions will come into effect on the ground when the respective State governments pass orders to implement them.

As per law, the State governments are bound to implement directions of the CAQM and the Commission can initiate legal action against officers not following its directions.

The GRAP was first notified by the Ministry of Environment, Forest and Climate Change in January 2017, pursuant to Supreme Court directions to provide for graded measures corresponding to different levels of air quality.

Initially, the GRAP was implemented by the EPCA (Environmental Pollution (Prevention and Control) Authority), which was formed in 1998 by the Supreme Court and was responsible for the management and coordination of air pollution in Delhi and the NCR.

But the EPCA was not backed by a statute, and experts had raised the issue that it lacked the teeth to act against authorities that did not follow its orders.

In 2021, the CAQM replaced the EPCA, and since then it has been implementing the GRAP. Over the years, at times after coming under attack from the Supreme Court, the GRAP has gone through multiple revisions, making it more stringent.

What now?

Most of the actions under Stage 1 are directing authorities to strictly implement existing norms related to air pollution.

The four stages of GRAP are: Stage I - 'Poor' Air Quality (Delhi AQI ranging between 201-300), Stage II - 'Very Poor' Air Quality (Delhi AQI ranging between 301-400), Stage III - 'Severe' Air Quality (Delhi AQI ranging between 401-450), Stage IV - 'Severe +' Air Quality (Delhi AQI > 450).

A higher AQI indicates worsening air quality.

Though the GRAP can be imposed any time during the year, the higher stages of GRAP are mostly invoked during winter as air quality deteriorates.

Under GRAP 1, the CAQM has directed Delhi and its neighbouring States to ensure that diesel generator sets are not used as a regular source of power supply, among other measures.

It's under Stage III and Stage IV of the GRAP that more severe actions kick in.

For instance, under Stage III, the CAQM announces a ban on construction and demolition activities in Delhi-NCR and a ban on the plying of BS III petrol and BS IV or below emission standard diesel Light Motor Vehicles (LMVs) or cars in Delhi and in the High Vehicle Density (HVD) districts of Gurugram, Faridabad, Sonipat, Ghaziabad and Gautam Buddh Nagar.

Similarly, under Stage IV, the ban on construction becomes even more severe with fewer exemptions, and there is a ban on the entry of trucks into Delhi, except for trucks that are BS-VI, electric, CNG, and LNG. Other measures are also taken under both the stages.



Winter woes: People walk along Kartavya Path as haze envelops Rashtrapati Bhavan; every winter, air pollution spikes in Delhi-NCR and many parts of the larger Indo-Gangetic Plains, mainly due to meteorological factors. ANI

Does GRAP actually help?

The GRAP aims to cut down the sources of pollution for a short time when the air pollution is very high, so that the air quality improves, but it is not a long-term strategy.

According to an analysis released on Thursday (October 22, 2026) by the Centre for Research on Energy and Clean Air (CREA), a Finland-based non-profit research organisation, there has been "no consistent" improvement in Delhi's winter air quality over the past four years, despite the imposition of GRAP.

"Delhi's winter AQI (Air Quality Index) has fluctuated over the past four years, with no consistent improvement. The mean AQI rose from 280 in winter 2022-23 to 304 in 2023-24, fell to 285 in 2024-25 and increased again to 295 in 2025-26. The median AQI followed a similar pattern, rising from 294 to 324, falling to 289 and then increasing to 305," CREA said in a statement.

Delhi's winter air quality has not followed a consistent downward trend since the 2022 revision of GRAP, and the mean and median AQI for the latest winter were both higher than in 2022-23, said Manoj Kumar, an analyst at CREA.



Vehicles pass as an anti-smog gun sprays water to curb air pollution in central Delhi, where GRAP-I measures are in effect, in New Delhi on Tuesday. ANI

What is different this winter?

In addition to GRAP, the Delhi government in June this year issued a gazette notification of a set of restrictions for winter months. But unlike GRAP, these restrictions will come into play irrespective of the air quality at a particular time.

For instance, as per the June 23 notification, from November 1 to January 31, there will be a ban on the plying of all non-Delhi-registered motor vehicles below BSVI emission standards in the national capital. However, electric, CNG, emergency, and sovereign duty vehicles are exempted from it.

Similarly, as per the notification, from November 1 to February 28, existing parking charges at authorised parking sites will be doubled to discourage use of private vehicles, except for parking spaces owned and managed by the Delhi Metro Rail Corporation (DMRC).

But if there is any conflict between the GRAP orders and the June 23 notification by the Delhi government, the GRAP will take precedence.

With the additional restrictions imposed by the Delhi government, this winter is set to see one of the highest levels of government action in the recent past in Delhi, at least on paper.

THE GIST

- GRAP imposes increasingly stringent restrictions as Delhi's air quality worsens, from 'Poor' to 'Severe+', but its measures are short-term
- Delhi's winter air quality has shown no consistent improvement over the past four years despite GRAP, with the mean AQI rising from 280 in 2022-23 to 295 in 2025-26, according to CREA
- The Delhi government has introduced additional curbs, including a ban on non-Delhi-registered vehicles below BS-VI standards and doubled parking charges at authorised sites, regardless of air quality levels

KEYWORD

How neoliberalism turned the world against its young

The rise in youth unrest stems from failures of neoliberal capitalism, which has weakened public welfare and worsened unemployment; as falling wages undermine consumption and investment, young people face fewer jobs, rising education costs and an economic system that prioritises profits

Rohit Azad
Indranil Chowdhury

France is in turmoil. And Albania. And Spain. Even India. As were Bangladesh or Sri Lanka earlier. The list goes on. While the specific demands may vary, one thing is common across these mobilisations: the young are on the streets. Why are the young angry the world over? In this piece, we look at the first world, in particular the G7 bloc, to understand why indeed.

France is a good example to understand what is happening in the global North. Emmanuel Macron, the antagonist according to the students in the French protests, has been in power since May 14, 2017.

Over his almost a decade-long reign, France has witnessed severe economic problems – high unemployment, low growth, rising cost of living, falling real wages and severe cutbacks in public expenditure. It is not a happenstance, therefore, that the spark that lit this protest was the lack of resources and staff at the Lycée Antoine-de-Saint-Exupéry in Créteil (a suburb of Paris).

At the heart of it all is the neoliberal economic model, which aggressively enforced reducedpublic expenditure in general, and welfare spending in particular. Whether it's Macron in France or Biden/Clinton in the U.S. or Starmer in the U.K., those championing this economic model have either been ousted or are on the brink of losing power.

Crisis of unemployment

The system has angered young people in particular. Although overall unemployment rates have remained high since the pandemic, youth unemployment rates in G7 countries are now almost twice as high, if not higher (Chart 1). Generation Z (Gen Z), which has taken to the streets, comprises those born between 1997 and 2012, putting them in the 14-29 age group today. That cohort overlaps with the age group 15-29 shown in the chart.

As is well known, unlike countries in the Global South, developed countries are ageing societies, with the share of young people in their populations declining over time. Yet, as their relative population share shrinks, youth unemployment continues to rise. While fewer young people enter the labour force each year, many more join the ranks of the unemployed. Clearly, the problem is not an excess supply of labour but a lack of demand for it.

What causes unemployment?

As is the case with most economics questions, this also has broadly two sides – orthodox and heterodox. The orthodox side believes unemployment has to do with high rigid wages, inflexible labour markets, lack of a skilled labour force, most of which puts the blame, directly or indirectly, on the working class itself for its misfortune.

The heterodox side, in sharp contrast, argues it is the decision of the capitalist class to employ or not to employ workers which determines unemployment in the economy.

Consider a typical labour market – demand for labour comes from the capitalists; supply of it comes from the workers. Capitalists would like to employ as many as they could to maximise their profits. The lower the wage rate, the higher the profits, and so the demand for labour rises as wages fall (downward-sloping demand curve in graph 1). The supply curve, on the other hand, is upward sloping because as the wages rise, more workers join the labour force. The intersection of the two (point E) represents full employment – those who seek work at that wage get it.

When the economy gets stuck at point A – demand for labour (point A) is less than its supply (point B) – the economy has involuntary unemployment (equal to the gap AB). But what stops the economy from reaching point E?

Orthodox economics would tell us the wages are way too high, for which either the labour unions or rigid labour market legislation (against 'hire and fire', for example) are responsible. To be sure, there are other theories like the 'efficiency wages' which explain why the economy remains stuck at A.

The heterodox approaches argue that labour demand depends on the level of demand for goods and services in the economy. Demand, under 'pure' capitalism, consists of investment and consumption (see (A) in box 1). Let's assume, for simplicity, workers consume all their wages and capitalists save their entire profits. Consumption demand would then depend on employment (wage bill), so it cannot be the cause of unemployment. The real cause, therefore, is investment. The less the capitalists decide to invest, less will be the total induced demand for goods and services, and the lower will be the need to employ workers.

It is the capitalists, not the workers, who are responsible for unemployment.

But why can the State not fill the demand gap through fiscal expansion and ensure perpetual full employment? After all, it is a win-win both for the workers and capitalists – those willing to work get work and profits rise with fiscal expansion (see (B) in box 1). That was the Keynesian route out of the capitalist crisis.

In his famous essay on 'Political Aspects of Full Employment', Michal Kalecki argued against this naïveté of Keynesianism because "under a regime of permanent full employment, 'the sack' would cease to play its role as a disciplinary measure".

The rise of the neoliberal world order in response to the 'New Deal' of the 1950s and the 1960s mirrors Kalecki's statement word for word.

Rise of neoliberalism

What was the 'New Deal'? In response to the Great Depression of the 1930s – the deepest and the longest unemployment and profitability crisis capitalism has ever faced – a new world order emerged which argued for larger governments to push the economy towards point E. This created what is called the Golden Age of Capitalism (1945-1973), which recorded the highest ever rates of growth in the global North (see chart 2). But it came along with a class alignment in favour of the working class, whose shares in the total economic pie kept rising throughout the first world (see chart 3). The 'sack' was losing its role as a disciplinary measure.

A counter-revolution, led by the economist Milton Friedman, broke out, which sought to bring the 'sack' back. Inflation that broke out between 1973 and 1980 – the pretext of this countercharge – was cost-push (oil price shocks), and had absolutely nothing to do with larger governments. Despite that, the monetarist countercharge argued that government meddling with a 'natural' state of affairs under capitalism caused this unprecedented inflation. Capitalism, on its own, produces full employment and so let the markets do their job, it was argued and that governments had no business to be in business.

Ronald Reagan in the U.S. and Margaret Thatcher in the U.K. put the monetarist doctrine into practice, which has since come to be known as the neoliberal world order. With the withdrawal of the welfare state, large-scale union busting and 'flexible' labour markets, the balance of class forces was restored in favour of the capitalists. The share of the working class in the economic pie nose-dived and the advances of the Golden Age were reversed (see chart 3).

Crisis of underconsumption

A fall in the wage share, however, comes with its own problems. What makes sense for an individual capitalist – a fall in wage cost – harms the capitalist class as a whole, as far as profits are concerned. It happens for two different reasons. One, a dollar's fall in wages means an equivalent fall in consumption demand, which means a fall in profits of capitalists selling these consumer goods. Two, such a fall in consumption leads to a second-degree fall in investment in the next period because why would the capitalists build more factories if the existing ones become idle? Karl Marx called it the crisis of underconsumption. Chart 3, seen together with Chart 2, is this crisis in action.

This fall in growth creates a crisis of unemployment, which is further aggravated by capital's bid to increase its profit margins by replacing workers with machines. It is a double whammy for the youth – low growth, which is labour-displacing. Artificial Intelligence (AI) could potentially aggravate this crisis.

Last but not the least, when workers are led to believe that their lack of skills is to blame for their unemployment, they turn to education to upgrade their skills. But neoliberalism ensures that education is turned into a money-making industry worldwide.

The cost of education has been skyrocketing under the neoliberal phase (see chart 4).

Getting a degree becomes more expensive, and those who manage to get it don't necessarily land themselves remunerative jobs. Who can blame the youth, therefore, for resisting this system rigged against them by design?

(The authors are a part of the DevMac (Development Macroeconomics) network, which seeks pluralism in Economics)

A burden on the young

Neoliberalism has reversed past economic gains, leaving the young with fewer opportunities and greater insecurity

Chart 1: Wrath of the unemployed Gen Z

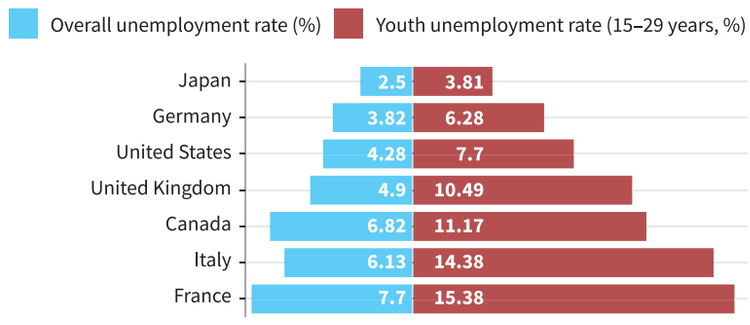


Chart: Azad & Chowdhury. Source: Authors' calculations. International Labour Organisation (ILO), International Labour Organisation Statistics (ILOSTAT)

Chart 2: Growth rates for the G7 bloc (Period-wise CAGR in %)

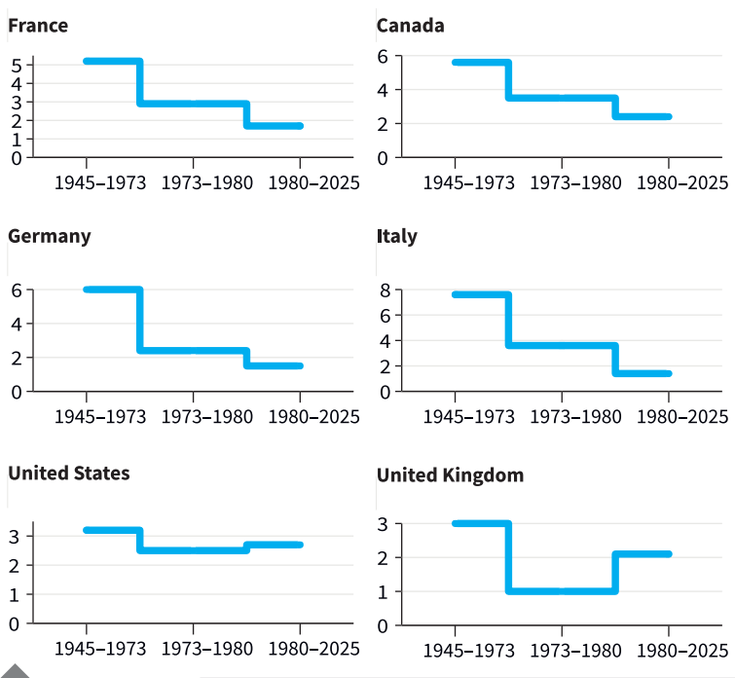
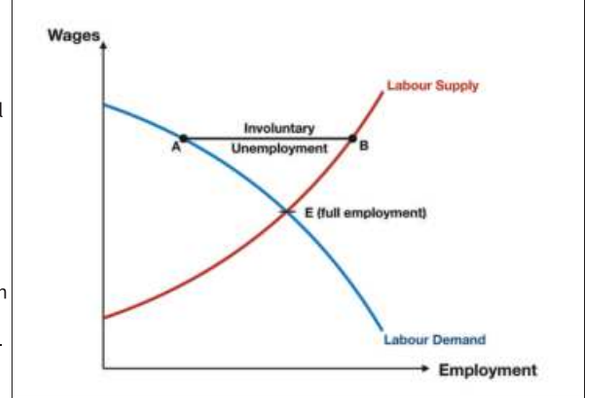


Chart: Azad & Chowdhury. Source: Authors' calculations. U.S. Bureau of Economic Analysis (BEA). Federal Reserve Bank of St. Louis (FRED). Institut national de la statistique et des etudes economiques (INSEE). Federal Statistical Office (Destatis). Italian National Institute of Statistics (ISTAT), Istat-Data. Cabinet Office, Economic and Social Research Institute (ESRI). Statistics Canada. World Bank, World Development Indicators. OECD, National Accounts at a Glance



(A) A "pure" capitalist economy:

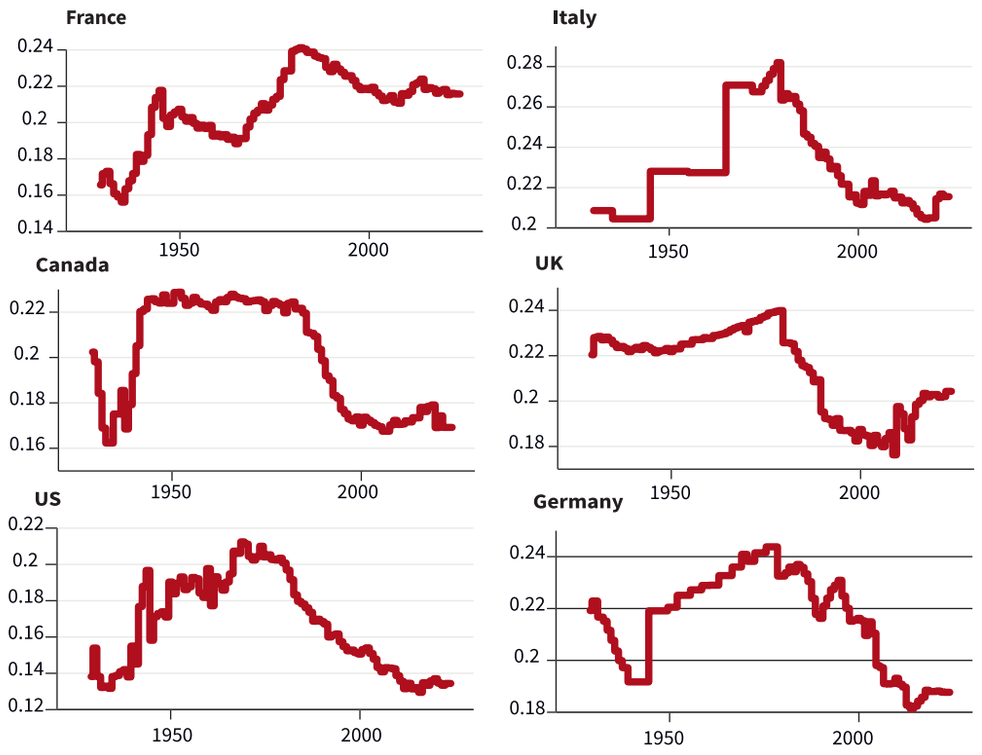
$$\frac{\text{Wage bill}}{\text{Wage x Employment}} + \text{Profits} = \text{Consumption} + \text{Investment}$$

(B) With state intervention:

$$\text{Wage bill} + \text{Profits} = \text{Consumption} + \text{Investment} + \text{Government Expenditure}$$

$$\text{Profits} = \text{Investment} + \text{Government Expenditure}$$

Chart 3: Share of the poorest half in the total income of G7 countries



Some data points for Italy and Germany are not available, so a connecting line through the available data points has been generated for uniformity Chart: Azad & Chowdhury. Source: World Inequality Database

Chart 4: Rising cost of education in G7

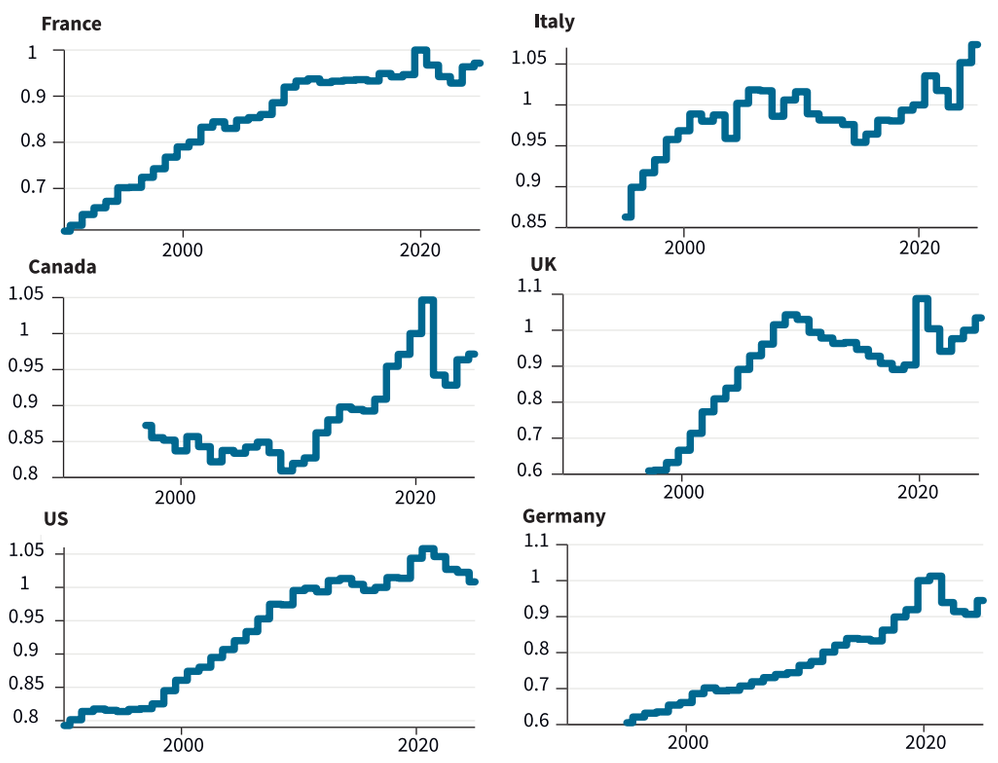


Chart: Azad & Chowdhury. Source: Authors' calculations. U.S. Bureau of Economic Analysis (BEA), NIPA, republished by FRED. Statistics Canada. Eurostat. OECD National Accounts. World Bank, World Development Indicators. Economic and Social Research Institute, Cabinet Office of Japan. UK Economic Accounts (UKEA).



In rage: France has witnessed some of the worst economic problems — high unemployment, low growth, rising cost of living, falling real wages, severe cutbacks in public expenditure. AFP

From Page One

SC asks if ECI has powers to punish voters

"First and foremost, in Bihar SIR, the ECI had not undertaken 'logical discrepancy' as a category. We had permitted you in West Bengal as an interim arrangement... We have not finally adjudicated on the legality of 'logical discrepancy'," Justice Bagchi spoke for the Bench.

Justice Bagchi asked how voters, included in the 2003 electoral roll after an intensive revision, could be purged in the name of 'logical discrepancy' in 2026.

Petitioners, activists Anjali Bhardwaj and Amrita Johri, represented by advocates Prashant Bhushan and Neha Rathi, submitted that the ECI was compelling voters to misrepresent themselves as new applicants, an offence which attracts imprisonment.

"You cannot touch these Forms. Form 6, Form 7 [for objection to proposed inclusion/deletion of names in the existing electoral roll], and Form 8 [application for shifting of residence] are cast in stone... You cannot make changes to them," Justice Bagchi said.

Mr. Bhushan said the ECI's approach to voters in the Delhi SIR was short-sighted.

"It ran like 'you were absent at your address when we came with the enumeration form, you are deleted from the electoral roll'. Of the 1,45,10,299 electors in Delhi at the start of the SIR, 47,56,722 were marked as having not returned their enumeration forms on account of being absent, shifted, dead or duplicate entries," he submitted.

The court asked the ECI whether a person who had temporarily moved out of Delhi for a month-long assignment, and missed filling out his enumeration form, would be considered as 'permanently shifted'.

"If a voter has moved residence from Vasant Kunj to Sarita Vihar, is it not enough for him to simply disclose his voter details to continue in the electoral roll," Chief Justice Kant asked the ECI.

Justice Bagchi said the requirement for voters to apply through Form 6 and Form 7 would arise only after the final voter list was published, and not during the draft stage.

"Why don't you allow the deleted voters to continue to submit their enumeration forms? There is a structural incongruity between your manner of conduct of SIR and the Form 6 and 7 regime. Why should voters be asked to apply with Form 6, they can just put in their enumeration forms belatedly... Keep it open-ended. Don't stick to any timelines for voters," Justice Bagchi addressed the ECI.

Chief Justice Kant asked why the ECI had to rush the SIR in Delhi, with no elections around. The last date for filing claims and objections for inclusion/deletions in the Delhi electoral roll is October 30.

Mr. Naidu said voters were never 'deleted' as they could return to the electoral roll at any time before the filing of nominations for an election.

India raises concern over U.S. support for Pakistan's F-16 jets

Rajnath Singh discusses cross-border terrorism with Hegseth; Secretary of War reaffirms support to combat issue; they review defence relations

Saurabh Trivedi
NEW DELHI



Defence Minister Rajnath Singh during a meeting with United States Secretary of War Pete Hegseth on Wednesday. PTI

Defence Minister Rajnath Singh on Wednesday raised India's concerns over Washington's support package for Pakistan's F-16 fighter aircraft with U.S. Secretary of War Pete Hegseth during a bilateral meeting on the sidelines of the ASEAN Defence Ministers' Meeting-Plus (ADMM-Plus) in Cebu, Philippines.

The two leaders also discussed the threat of cross-border terrorism, with Mr. Hegseth reaffirming Washington's support to India in combating the menace.

According to the Ministry of Defence, the meeting provided an opportunity to review progress under the India-U.S. Major Defence Partnership, including the implementation of foundational security agreements and avenues to enhance operational readiness, technology sharing and defence industrial integration.

Mr. Singh emphasised the need for a zero-tolerance approach towards countries that support, train or finance terrorist organisations. He expressed appreciation for the consistent support extended by the U.S. in India's fight against cross-border terrorism.

Mr. Singh also conveyed to Mr. Hegseth that any transfer of advanced military technology to Pakistan could ultimately result in the technology reaching a third country, raising concerns over its wider proliferation and implications for regional security.

The two Ministers also reviewed the ten-year India-U.S. Defence Framework and discussed strengthening maritime security cooperation, military-to-military exchanges and resilient defence supply chains.

Separately, Mr. Singh held talks with Japanese Defence Minister Shinjiro Koizumi, reviewing progress in bilateral defence cooperation since their meeting in August. The discussions explored joint development in naval shipbuilding and design.

DPDP Act does not infringe on press freedom: Vaishnaw

The Hindu Bureau
NEW DELHI



Ashwini Vaishnaw

Union Information Technology Minister Ashwini Vaishnaw on Wednesday said the Digital Personal Data Protection (DPDP) Act did not infringe on press freedom, asserting that the Constitution guaranteed journalistic independence.

"The journalistic freedom is guaranteed by the Constitution. That freedom is absolutely available. There is not even an intention to affect the journalistic freedom," Mr. Vaishnaw said. He invited editors and media professionals to engage in an open dialogue with the government on concerns surrounding the law.

The remarks came amid concerns from news publishers that the law could make investigative reporting more difficult. They have argued that the absence of a specific exemption for journalism could affect reporting on matters of public interest, especially investigations involving personal information collected without consent. They have also raised concerns about retaining information required as part of investigative reporting, and protecting confidential sources. The Minister said the privacy law "fully maintains the journalistic freedom the Constitution guarantees", adding that it was aimed at protecting personal data without undermining media freedom.

It is learnt that the government is also open to amending the law to explicitly safeguard journalistic activities, if necessary. "If hardcoding is required, we can hardcode it; it only requires an amendment... There is no problem in that because our intent is clear," a top source said. The source added that the government was willing to discuss the concerns and that any confusion could be addressed through an amendment in Parliament.

Mr. Vaishnaw said the DPDP Act was introduced following a Supreme Court judgment recognising the right to privacy.

(With agency inputs)

Japan-northeast India link to be strengthened: Centre

Kallol Bhattacherjee
NEW DELHI



Randhir Jaiswal

India has a "very strong partnership" with Japan, and bilateral projects involving the northeastern States will be "further strengthened", Randhir Jaiswal, official spokesperson of the External Affairs Ministry, said on Wednesday.

The comments came hours after Bangladesh Prime Minister Tarique Rahman and his Japanese counterpart, Sanae Takaichi, agreed to "enhance connectivity in the region", and signed agreements that will pave the way for additional Japanese assistance to Bangladesh's Matarbari deep sea port, which will connect with northeast India.

"We have very strong partnership with Japan. We also have several initiatives involving northeast [India] and Japan, and we would like to further strengthen these linkages," Mr. Jaiswal said. The Japanese language was being taught to the working population in the northeastern States as part of these initiatives, he added.

"It will also give a boost to the Japan-India economic partnership here in India as well," Mr. Jaiswal said. Northeastern States were also receiving assistance from Japan as part of that larger bilateral framework, he said.

Japan and Bangladesh have signed multiple agreements during Mr. Rahman's ongoing visit to Tokyo, which includes further development of the industrial hub centred around the Matarbari deep sea port. Mr. Rahman met Ms. Takaichi and representatives of leading business houses during the visit.

Sources from Japan maintain they envisage the linking of the Matarbari port with India's northeast to ensure optimal usage of the industrial hub and the first deep sea port in Bangladesh, which currently depends on ports in Malaysia and Singapore for handling big vessels.

A joint statement issued during Mr. Rahman's visit has included the agreement on the Matarbari port under the "connectivity" section. "The two Prime Ministers reaffirmed their commitment to steadily advance cooperation aimed at further strengthening connectivity in the region," it said.

Data shared in masked form, says ULIP amid complaints over sale of vehicle information

Jagriti Chandra
NEW DELHI



The Road Transport Ministry sought an audit after finding that some entities were extracting data in large volumes from ULIP. FILE PHOTO

After the Ministry of Road Transport sought an audit of users accessing the Centre's logistics portal following complaints that multiple apps and websites were selling vehicle information, the Unified Logistics Interface Platform (ULIP) rejected the allegations of its misuse. It said that all data were masked and called for broader scrutiny of the platforms involved.

"ULIP relays information in the format received from the source API. For example, vehicle owner names, chassis and engine numbers retain their masking, and mobile number field remains null in both the source response and the response provided to users. ULIP does not independently generate, unmask or enrich the underlying data," ULIP wrote in response to a story in *The Hindu*.

The Ministry sought an audit after finding that some entities were extracting data in large volumes from ULIP with some pulling out as many as 50 lakh records a month. The Road Ministry shares vehicle information with multiple government agencies and private firms from its portals such as VAHAN, SARATHI, e-Challan and PUCC portals.

ULIP said it uses sufficient safeguards that include screening of entities that have access to its portal, their proposed use cases, binding terms and conditions, IP whitelisting, phased onboarding and continuous monitoring of API usage. This framework complies with the Road Ministry's Policy for Data Sharing from the National Transport Repository which prescribes authentication, secret keys, masking of personally identifiable information (PII) and defined approval mechanisms, ULIP added.

Referring to multiple apps selling vehicle information data in unmasked form, ULIP said there needed to be a "scrutiny of wider ecosystem of platforms and channels through which such information is being made available".

The Road Ministry has written to the Indian Computer Emergency Response Team (CERT-In), the government's nodal agency for investigating cybersecurity incidents, seeking a forensic audit of VAHAN, SARATHI, e-Challan and PUCC portals, which hold a range of vehicle-owner information.

It has also sought the intervention of the Ministry of Electronics and Information Technology (MEITY) for a review of its data sharing policy and suggest ways to strengthen it. The Road Ministry also plans to audit various enforcement agencies that access its portals for data.

Regulator cuts prices of 543 anti-cancer medicines

Bindu Shajan Perappadan
NEW DELHI

The National Pharmaceutical Pricing Authority (NPPA) on Wednesday revised downward the maximum retail price of 543 anti-cancer medicines of different brands, effective November 9.

On October 17, the NPPA had announced Trade Margin Rationalisation (TMR) of 137 non-scheduled anti-cancer drugs in public interest to address excessive trade margins, make cancer treatment more affordable, and improve access to anti-cancer medicines.

Meanwhile, Jan Swasthya Abhiyan India (JSA India), an organisation working for accessibility of medicines, said that controlling trade margins alone is insufficient. Life-saving cancer treatment must be affordable based on medical need, not a patient's ability to pay.

"For example, the reported MRP of one brand of palbociclib fell from approximately ₹40,421 to ₹2,720 per pack, while thiotepa fell from approximately ₹80,072 to ₹7,150 per vial. Such drastic reductions raise serious questions about the pricing practices of pharmaceutical companies," the group said.

Union Cabinet gives nod to ₹7,396-crore panchayat scheme

Sobhana K. Nair
NEW DELHI

The five-year scheme will have a Central share of ₹4,869.48 crore and State share of ₹2,526.07 crore

The Union Cabinet on Wednesday approved the Pradhan Mantri Viksit Gram Swaraj Yojana (PM-VGSY), a ₹7,395.55-crore scheme aimed at strengthening the planning and governance capacities of gram panchayats over the next five years.

The scheme will be implemented from 2026-27 to 2030-31 with a Central share of ₹4,869.48 crore and a State share of ₹2,526.07 crore. The government said around 32 lakh elected representatives, panchayat functionaries and other stakeholders across 2.55 lakh gram panchayats and traditional local bodies would be trained under the programme.

The new scheme is an upgraded version of the Rashtriya Gram Swaraj Abhiyan, with a greater emphasis on "holistic" development, as opposed to the existing practice of providing separate training modules for individual schemes. The idea is to offer "digital handholding" for the panchayat representatives to help them with evidence-based planning. Though the government refers to the 32 lakh representatives as "beneficiaries," it does not offer any stipend.

The scheme seeks to position gram panchayats as the focal point of local governance and development planning. It envisages preparing 2.55 lakh Gram Panchayat Development Plans (GDPDs) through the eGramSwaraj portal, enabling village-level priorities to align with Central and State government schemes and resources.

Awareness drives will be conducted across the country as part of the programme, with each campaign expected to reach at least one lakh citizens. A key element of PM-VGSY is the creation of a common digital platform that will integrate information on government schemes, beneficiaries, community assets and available resources. This will help panchayats identify development gaps, improve departmental coordination, and monitor implementation more effectively.

Railways to turn off charging points in trains during night

V. Geetanath
HYDERABAD

Train passengers will no longer be allowed to keep their mobile phones charging overnight, with the Railway Board issuing strict instructions to all zonal railways to switch off charging points between 11 p.m. and 5 a.m.

In an official communication, the Board has directed railway authorities to conduct surprise inspections in coaches during overnight journeys. Both officers and supervisory staff have been asked to ensure strict compliance with the directive, with no exceptions permitted under any circumstances.

Senior railway officials said a circular mandating the shutdown of mobile charging circuits during night travel had been issued more than a decade ago. However, the rule was apparently not being enforced consistently on some trains.

The Railway Board's renewed insistence on implementing the measure follows a recent incident in the South Eastern Railway zone. Railway staff reportedly failed to switch off charging points during the prescribed hours, leading to a power bank explosion while being charged.

However, some officials believe the focus should be on preventing the charging of power banks during the six-hour overnight period rather than restricting charging altogether.

They also argued that the number of charging points should not be increased indiscriminately.

SC agrees to hear AITUC challenge to definition of 'industry' in new law

Krishnadas Rajagopal
NEW DELHI

The Supreme Court agreed on Wednesday to hear a petition filed by the All India Trade Union Congress (AITUC) Tamil Nadu challenging the definition of 'industry' under the Industrial Relations Code (IRC) 2020 as "deliciously vague", especially when the new law exempts a wide array of institutions from charitable to sovereign functions of the government.

The IRC affords legal protections to workers who are employed in institutions which come under the ambit of 'industry'. Section 2(p) of the IRC defines industry as any institution engaged in a "systematic activity carried on by co-operation between an employer and worker for the production, supply or



distribution of goods or services with a view to satisfy human wants or wishes".

Appearing before a Bench headed by Chief Justice of India Surya Kant, senior advocate Gopal Sankaranarayanan and advocate Aarthi Rajan, for the trade union, referred to the exceptions to 'industry' listed in the Code.

Mr. Sankaranarayanan submitted that the exceptions to 'industry' included institutions wholly or substantially engaged in charitable, social or philanthropic service to "any activity of the appropriate government relatable to sovereign functions" to "any other activity as may be notified by the Central government".

"Is there a problem with the definition of 'industry'?" Chief Justice Kant asked.

The senior counsel said the definition in the Code did not meet the requirements of the 'triple test'. A 1978 seven-judge Constitution Bench judgment, authored by Justice V.R. Krishna Iyer, in the *Bangalore Water Supply and Sewerage Board v. A. Rajappa* case, had laid down conditions that any activity which was systematic or organised, operated on the basis of employer-employee cooperation and dealt in the production of goods and services to satisfy human wants came within the ambit of 'industry'. The judgment had brought even hospitals, educational institutions and municipalities within the fold of 'industry'.

In August 2026, a nine-judge Bench headed by Chief Justice Kant had reviewed the interpretation given by Justice Iyer in 1978 to the definition of 'industry' under Section 2(j) of the Industrial Disputes Act, 1947. However, the nine-judge Bench had been preceded by the coming into force of the 2020 Code, which largely made the reference to it infructuous.

"The future of 'industry' in Section 2(p) of IRC is not burdened by the interpretation of 2(j) in the Industrial Disputes Act," Chief Justice Kant had observed.

‘India requires $220 billion for green power transition by FY30’

Saptaparno Ghosh
NEW DELHI

India’s renewable energy sector will need $220 billion in investment between FY27 and FY30, spanning generation, transmission, manufacturing and emerging green technologies, Santosh Sarangi, Secretary, Union Ministry of New and Renewable Energy, said.

Another senior official told reporters at the curtain-raiser for the Bharat Renewable Energy Summit & Expo 2026 that solar energy would account for $70-75 billion of the requirement followed by transmission infrastructure at $40-45 billion, wind energy at $25-30 billion and green hydrogen at $20-25 billion. Manufacturing will require another $15-20 billion, the official said.

The sector attracted about $64 billion in private investment between July 2024 and June 2026, including $6.2 billion in foreign direct investment. The Centre infused $6 billion to support renewable energy development.

Mr. Sarangi said Indian green hydrogen developers got orders to supply 5.5 lakh tonne until 2026. The Ministry is developing a framework for geothermal exploration rights similar to hydrocarbon sector.

New genetic tools are breathing fresh life into India's staple crops

As new tools like apomixis become more widely available and private companies shift focus to Indian seeds, it is as important to think about how our system will adapt to these changes as it is to focus on investing in fundamental research in plant genetics, physiology, and host-pathogen interactions

K. VijayRaghavan



India's food security largely depends on rice and wheat, which take up more than half our farmland and provide most of the calories we eat. THOMPSON LE/UNSPLASH

Technology is rapidly changing the way we cultivate rice and wheat, and India has the opportunity to lead the pack if we restore the attraction of fundamental research in plant sciences to our students. The shrillness of anti-GMO voices has driven aspiring biology researchers to other areas of life sciences, which use much the same methods. We need to bring them back to plant research, too.

India's food security largely depends on rice and wheat, which take up more than half our farmland and provide most of the calories Indians eat. Scientists and breeders designed today's wheat and rice varieties at a time when the consequences of excess fertilizer use on soil were not sufficiently anticipated, when pesticides were used to hit insects and other pests with a metaphorical hammer, and when water and electricity were abundant or heavily subsidised.

Today, however, climate change, extreme weather events, droughts, and floods are more immediate concerns. This together with demands for sustainable and moderated use of pesticides and fertilizers, but without compromising yield, has created a high-bar of demand in a competitive world. In addition, science and technology are but one component of solutions that also involve politics, economics, and global supply chains.

Public institutions

The supply of seeds to farmers, for crops such as rice and wheat, has had a strong public-sector presence. The National Seeds Corporation (NSC) was established in 1963. The NSC partners with State seed corporations and State agricultural universities to ensure reliability in the early stages of seed supply. The Indian Council of Agricultural Research (ICAR) laboratories and State agricultural universities ensure supply of 'breeder seeds', certified to be of the highest quality. These are then amplified by partners, which include private companies to generate 'foundation seeds', which are distributed.

For wheat, public institutions dominate the supply of certified seeds, with private players being present in only about 5% of the market. However, their share is growing rapidly, at about 8% each year. In the case of rice, private companies dominate the market for hybrid seeds, with public institutions mainly supplying farmers with small-holdings.

In India, public institutions play a large role in wheat and rice, unlike with maize and cotton, where private companies dominate. The new varieties they develop are available to any registered seed company, public or private. The government has a system in place to make and distribute these new varieties widely, cheaply, and fairly – a system which works reasonably.

India took a big step forward in 2025 in using new gene-editing technology to improve rice, and became the first country to allow farmers to grow rice that had been edited using this technology. The scientists also grew these rice strains in fields, and the approved seeds are now being given to farmers.

Two types of rice, DRR Dhan-100 and Pusa DST Rice-1, were developed by Indian researchers using the CRISPR gene-editing technology. Researchers tested these new rice types widely in 2023-2024. The advances were significant. One variety DRR Dhan-100 took 130 days to mature, compared to 150 days of the baseline strain. It produced close to 20% more rice, at about 5.4 tonnes per hectare, and reduced water use and labour costs. Pusa DST Rice-1 was developed to grow reasonably in salty or alkaline soils. Its yield in these conditions was about 10-30% more than the baseline strains. Government announcements point to lowered water and fertilizer use and, consequently, lowered greenhouse gas emissions as well.

Tale of two varieties

The story behind these two strains is worth understanding for how fundamental research on rice flowering leads to application. The targeted gene in DRR DNAn-1 encodes a product that degrades cytokinin, the hormone driving spikelet formation on a rice panicle. This process caps how many grains a plant can produce.

ICAR's CRISPR edit partially reduces, but does not eliminate, this enzyme's activity, letting cytokinin accumulate just enough to boost spikelet and grain number without the stunted growth seen in complete knockouts. In 2005, Motoyuki Ashikari's team in Japan had already traced a grain-number trait to a natural small deletion in the corresponding gene in a high-yielding wild rice relative, and ICAR recreated that two-decade-old discovery by design, in weeks rather than through years of conventional breeding.

The PUSA DST-1 story is equally interesting. In 2020, a team at ICAR-Indian Agricultural Research Institute, led by Viswanathan Chinnusamy, used the CRISPR-Cas9 gene-editing tool to delete a stretch of a rice gene called *OsDST*, in a particular strain, producing plants with broader leaves, fewer stomata, and sharply better survival under salt stress (over 65% versus just 20% in unedited plants).

This engineered mutant reproduced the effects of a naturally occurring chemical mutation found years earlier in a completely different rice subspecies, confirming that disabling *DST* itself rather than some editing side-effect of the edit had conferred the stress tolerance. Pusa DST Rice 1, released five years later, is the commercial maturation of this already published laboratory mutant, illustrating a journey from a characterised gene to seed ready for farmers to use.

In both cases, the residues of the gene-editing tools were not detected in the final strains. Thus, while genetic engineering tools were used, the results are equivalent to those that would come from conventional mutagenesis and breeding, only much faster and more accurate.

Work on apomixis

Scientists at public institutions conducted the work on the rice, and the resulting seeds will not be tied up in restrictive patents or licensing agreements. The developers have also made sure that the technology used to create it will not be a bottleneck for public breeding programs.

Other innovative approaches are now on the horizon and will likely be deployable in the near future, led by a method called apomixis. It was first successfully demonstrated in rice by Venkatesan Sundaresan and his colleague Imtiyaz Khanday at the University of California, Davis, in 2018. In apomixis, plants are engineered to bypass the normal process of sexual reproduction and fertilisation. This allows desirable traits of the parent plant, known as hybrid vigour, to be preserved indefinitely, rather than being lost in the next generation. The efficiency of this method has improved significantly, with success rates reaching around 90% in rice. Various research groups are now exploring apomixis in other crops, such as maize and sorghum.

Sundaresan's work on plant development earned him a share of the prestigious Wolf Prize in agriculture in 2024, alongside Elliot Meyerowitz and Joanne Chory. In December 2025, Sundaresan and Khanday received the VinFuture Special Prize for their work on self-cloning crop technology, which has the potential to revolutionise agriculture.

There is still a big gap between the lab and the field. Tests in different environments are required to get regulatory approval and to set up systems to produce seeds in large quantities. Because adding foreign genes is not required to make apomixis work, many of these techniques could potentially follow a simpler regulatory process. The technology makes sense for small farmers. If they can save seeds from high-yielding crops and replant them without losing any quality, they will not have to keep buying new seeds every year. This is currently a big expense for many smallholder rice and wheat farmers.

Driving progress

India has not seen a gene-edited wheat released yet, but there are some conventional developments in the pipeline. For instance, the government recently introduced 184 new crop varieties across 25 different crops, including some climate-resilient wheat lines that are expected to reach farmers within the next three years. Hybrid wheat has been a challenge for breeders due to the complexities of wheat's floral biology. It is harder to hybridise than rice or maize. Despite this, hybrid wheat has been gaining traction in India, due in part to private companies like Mahyco and Nuziveedu.

Currently, the most advanced genetically engineered trait in wheat is called HB4, developed by an Argentine company called Bioceres Crop Solutions. This trait uses a gene from sunflowers to help wheat plants tolerate drought. This is not a new type of gene editing but a traditional genetically engineered trait. HB4 is already being grown commercially in Argentina, and on a smaller scale in Brazil and Paraguay.

In August 2024, the U.S. Department of Agriculture gave the green light for HB4 wheat to be grown in the country, making it the fourth to approve its production. However, it may take a few years before we see large-scale commercial planting in the U.S., since more field trials need to be done and the trait needs to be integrated into American wheat varieties. Additionally, export markets need to approve the trait as well.

Australia is taking a two-pronged approach to improving wheat yields, with two separate programmes running side by side. The Australian government has given the green light for field trials of HB4. This trial started in 2024. At the same time, the University of Adelaide is working on a project to engineer wheat and barley to produce higher yields, using genetic modification techniques. This project is set to run until 2029.

Meanwhile, an Australian breeding company called InterGrain has teamed up with a U.S. firm called Inari to try out CRISPR gene-editing. They are hoping to boost wheat yields by 10% and have already started field trials in the 2025 growing season. In a significant development, Australian Grain Technologies and China's Qihe Biosciences have announced that they will be working together on cross-border field trials of a new CRISPR-edited wheat variety. This is important as it shows that international collaboration is driving progress in wheat gene-editing, unlike some other crops like rice.

Reviving a tradition

India's approach to breeding wheat and rice is still largely controlled by the public sector, which means new techniques like editing genes without adding foreign ones can be made available to farmers.

As new tools like apomixis become more widely available and private companies start investing more in Indian seeds, it is as important to think about how our system will adapt to these changes as it is to invest in fundamental research in plant genetics, physiology, development and host-pathogen interactions.

We have had a great tradition here, and it should be revived.

(K. VijayRaghavan is the DAE-Homi Bhabha Chair, TIFR-National Centre for Biological Sciences, Bengaluru. vijay@ncbs.res.in)



India's approach to breeding wheat and rice is still largely controlled by the public sector, which means new techniques like editing genes without adding foreign ones can be made available to farmers. GETTY IMAGES



Salt marshes in India have been estimated to cover around 290 sq. km and support wildlife while stabilising coastal sediments. FILE PHOTO

How the humble salt marsh softens the ocean's fury

Vasudevan Mukunth

The boundary between land and sea is neither fixed nor clear. The soil does not suddenly stop and give away to a wall of water. Instead, the line is often a salt marsh, where ocean waves break and lose vim as they pass over the marsh's elevated surface and push against the stems of salt-tolerant plants.

Gujarat is home to India's largest coastal salt marshes, with smaller areas in Tamil Nadu, Andhra Pradesh, and Maharashtra, among some other places. One 2020 study in *Estuarine Coastal and Shelf Science* estimated they covered around 290 sq. km in all, and supported wildlife and stabilised coastal sediments. Unfortunately they also face significant threats from expanding aquaculture and coastal development.

As global warming continues, storms become ever stronger and water levels rise, these coastal zones are slowly being acknowledged for the formidable natural buffers they are. But they are being threatened as well, especially their plants, and what might happen if they are gone?

A 2014 study in *Nature Geoscience* used a wave flume – a long tank filled with water that uses a mechanical paddle to generate waves – to show that the plants of the salt marsh could break waves substantially even when their stems bent or broke.

Key findings

A new study in *Nature* by researchers at Wageningen University & Research in the Netherlands took the study a big step further. The team moved 306 sq. metres of a live salt marsh, down to the soil and roots, from the coast of the Dutch Wadden Sea to a 300-metre-long flume at the Deltares Research Institute, capable of creating the world's largest artificial waves. There, the marsh was tested at different depths against waves measuring 75 cm, 1.25 metres, and 2 metres high over many hours.

The findings were remarkable: the marsh's elevation fell only 7 mm but at the cost of the flora on the surface losing 75-80% of their biomass and having their stems become shorter by almost half. The researchers concluded the plants' dense network of roots had helped hold the sediments together. The team also found the plants cut the height of waves 75 cm high by 6-22% but those of waves 2 metres high only by up to 6%.

The team also combined their findings with maps of salt marshes worldwide and modelled the storm surges there. For once in a century storms – the kind climate change is rendering more common these days, like Cyclone Nisarga in 2020 – the team elucidated two kinds of outcomes.

Where the incoming waves are at most a metre high, the plants could be expected to cut the height of waves by over 10%, in some instances by as much as 50%. But if the waves got higher, the plants (alone) also became less effective at attenuating them.

Obviously these figures are not a guarantee for every coast everywhere. The study relied on a Dutch marsh dominated by one species of grass (*Elymus athericus*).

A different study published this year, in *Communications Sustainability*, also found that marsh species with more flexible stems often survived extreme waves better than stiffer ones. Nonetheless, the findings are important: they reinforce the ecological value of salt marshes, including the roots they still hold on exposed coasts.

mukunth.v@thehindu.co.in

For feedback and suggestions
for 'Science', please write to **science@thehindu.co.in** with the subject 'Daily page'